Lecture 10

13.11.25

FD methods for parabolic problems – convergence of the θ-scheme

Let us also briefly discuss the case (1D in space for simplicity)

$$u_t(t,x) \underbrace{- u_{xx}(t,x)}_{=Lu} = f(t,x) \quad \text{in } \underbrace{(0,T)}_{=I} \times \underbrace{(0,1)}_{=\Omega} \qquad (*)$$

(initial condition) $u(t=0,x) = u_0(x)$ in $(0,1)$

(Dirichlet boundary condition) $u(t,x=0) = 0 = u(t,x=1)$

$\rightsquigarrow$ we consider a discretization in space $\Omega_h = \{h, 2h, \dots, (N-1)h\}$ (equidistant), $h = \frac{1}{N}$ and

time $I_\tau = [0, \tau, 2\tau, \dots, M\tau]$ (equidistant), $\tau = \frac{T}{M}$

For fixed time t, we approximate $u_{xx}(t,x)$ as before (boundary nodes eliminated

$$u_{xx}(t,x) = \frac{u(t,x+h) - 2u(t,x) + u(x,x-h)}{h^2} + \mathcal{O}(h^2)$$

For fixed $x \in (0,1)$, we approximate $u_t(t,x)$ by a forward difference scheme

$$u_t(t,x) = \frac{u(t+\tau,x) - u(t,x)}{\tau} + \mathcal{O}(\tau) = D_t^+ u(t,x) + \mathcal{O}(\tau)$$

We will proceed step by step in time and collect all spatial values in one vector

$$u_h^k = \begin{pmatrix} u_1^k \\ u_2^k \\ \vdots \\ u_{N-1}^k \end{pmatrix}, \text{ where } u_i^k \text{ is an approximation of } u(k\tau, ih).$$

<u>Definition</u> (Theta scheme) For a real parameter $0 \le \theta \le 1$, we consider the following discretization of the parabolic problem $(*)$

$$D_t^+ u_h^k + \left(\theta L_h u_h^{k+1} + (1-\theta) L_h u_h^k\right) = f_h^k$$

for $k = 0, 1, \dots, M$ and a suitable approximation f_h^k of f. We speak of the <u>θ-scheme</u> and particularly consider

i) <u>Explicit/Forward Euler scheme</u>: $\theta = 0$ which componentwise reads

$$u_i^{k+1} = (1-2\gamma) u_i^k + \gamma (u_{i+1}^k + u_{i-1}^k) + \tau f(k\tau, ih), \quad \gamma = \frac{\tau}{h^2}$$

$$\left(\Leftrightarrow \frac{u_i^{k+1} - u_i^k}{\tau} = \frac{u_{i+1}^k - 2u_i^k + u_{i-1}^k}{h^2} + f(k\tau, ih) \right)$$

ii) <u>Implicit / Backward Euler scheme</u>: $\theta = 1$ which componentwise reads

$$(1+2\gamma)\, u_i^{k+1} - \gamma\,(u_{i+1}^{k+1} - u_{i-1}^{k+1}) = u_i^k + \tau f((k+1)\tau, ih), \quad \gamma = \frac{\tau}{h^2}$$

$$\left(\Leftrightarrow \frac{u_i^{k+1} - u_i^k}{\tau} = \frac{u_{i+1}^{k+1} - 2u_i^{k+1} + u_{i-1}^{k+1}}{h^2} + f((k+1)\tau, ih) \right)$$

iii) <u>Crank-Nicolson scheme</u>: $\theta = \frac{1}{2}$ which componentwise reads

$$(1+\gamma)\, u_i^{k+1} - \frac{\gamma}{2}(u_{i+1}^{k+1} + u_{i-1}^{k+1}) = (1-\gamma)\, u_i^k + \frac{\gamma}{2}(u_{i+1}^k + u_{i-1}^k) + \tau f(k\tau + \tfrac{\tau}{2}, ih), \quad \gamma = \frac{\tau}{h^2}$$

<u>Goal</u>: show consistency (and later convergence) of these schemes.

<u>Theorem (Consistency of θ-scheme)</u> The θ-scheme is consistent w.r.t. the system of maximum norms with

i) $\mathcal{O}(\tau + h^2)$, i.e., order 1 in time, order 2 in space, if $u \in C^{\overset{\text{time}}{2},\overset{\text{space}}{4}}$,

ii) $\mathcal{O}(\tau^2 + h^2)$, i.e., order 2 in time and space, if $\theta = \frac{1}{2}$ and $u \in C^{3,4}$.

<u>Proof</u> Recall: consistency in the time independent case $\| f_h - L_h R_h u \|$, i.e.,

how well does the continuous solution satisfy the discrete scheme

(already shown: $f_h - L_h R_h u = \mathcal{O}(h^2) \Rightarrow R_h L u = L_h R_h u + \mathcal{O}(h^2)$!)

$\rightsquigarrow$ analogy in the time dependent case: how well does the continuous solution satisfy the (discrete) θ-scheme?

We introduce the notation for the (spatial) restriction of the solution u by

$$R_h u^k = \begin{pmatrix} u(k\tau, h) \\ \vdots \\ u(k\tau, (N-1)h) \end{pmatrix} \in \mathbb{R}^{(N-1)}$$

From Taylor expansions of each component of this vector we conclude

- $$\frac{R_h u^{k+1} - R_h u^k}{\tau} = R_h\left(\dot{u}^k + \ddot{u}^k \frac{\tau}{2}\right) + \mathcal{O}(\tau^2) \quad \boxed{}$$

- $$L_h R_h u^{k+1} = L_h R_h (u^k + \tau \dot{u}^k) + \mathcal{O}(\tau^2) \quad \boxed{}$$

We can now insert u, respectively $R_h u^k$ in the θ-scheme to obtain

$$0 = \frac{1}{\tau}\left(R_h u^{k+1} - R_h u^k\right) + \theta L_h R_h u^{k+1} + (1-\theta) L_h R_h u^k - f_h^k$$

$$= R_h\left(\dot u^k + \ddot u^k \frac{\tau}{2}\right) + \mathcal{O}(\tau^2) + \theta L_h R_h (u^k + \tau \dot u^k) + \mathcal{O}(\tau^2)$$

$$+ (1-\theta) L_h R_h u^k - f_h^k$$

$$= R_h\left(\dot u^k + \ddot u^k \frac{\tau}{2}\right) + \theta \tau \underbrace{L_h R_h \dot u^k}_{= R_h L \dot u^k + \mathcal{O}(h^2)} + \underbrace{L_h R_h u^k}_{\overset{\shortparallel}{R_h L u^k + \mathcal{O}(h^2)}} - f_h^k + \mathcal{O}(\tau^2)$$

$$= R_h\underbrace{\left(\dot u^k + L u^k\right)}_{(1)} + \frac{\tau}{2} R_h \underbrace{\left(\ddot u^k + 2\theta L \dot u^k\right)}_{(2)} - f_h^k + \mathcal{O}(\tau^2) + \mathcal{O}(h^2)$$

(1) : means componentwise $u_t(k\tau, ih) - u_{xx}(k\tau, ih)$

$\rightsquigarrow$ since u is a continuous solution this yields $f(k\tau, ih)$

(2) : means componentwise $\frac{\partial}{\partial t}\left(u_t(k\tau, ih) - 2\theta\, u_{xx}(k\tau, ih)\right)$

$\rightsquigarrow$ IF $\theta = \frac{1}{2}$, this yields $\frac{\partial}{\partial t} f(k\tau, ih)$

Hence: if $\theta \in \{0, \frac{1}{2}, 1\}$ the definition for f_h^k reads componentwise (Taylor)

$$\theta = 0: \quad (f_h^k)_i = f(k\tau, ih)$$

$$\theta = 1: \quad (f_h^k)_i = f((k+1)\tau, ih) = f(k\tau, ih) + \mathcal{O}(\tau)$$

$$\theta = \tfrac{1}{2}: \quad (f_h^k)_i = f(k\tau + \tfrac{\tau}{2}, ih) = f(k\tau, ih) + \frac{\tau}{2} \dot f(k\tau, ih) + \mathcal{O}(\tau^2)$$

$\Rightarrow$ for $\theta \in \{0, 1\}$ expression (1) vanishes $\longrightarrow \mathcal{O}(\tau + h^2)$

for $\theta = \frac{1}{2}$ expressions (1) + (2) vanish $\longrightarrow \mathcal{O}(\tau^2 + h^2)$

For convergence, we need the following estimates.

<u>Theorem (estimates)</u> Let $M \in \mathbb{R}^{d \times d}$ be s.t. $M_{ii} > 0$, $M_{ij} \le 0$, $i \neq j$ and $\sum_{j=1}^{d} M_{ij} \ge 0$ for all $i = 1, \dots, d$ ("weakly diagonally dominant"). Then

i) $\|v\|_\infty \le \|(I_d + M) v\|_\infty$.

ii) If $0 < M_{ii} \le 1$, then $\|(I_d - M) v\|_\infty \le \|v\|_\infty$.

Proof

i) Let $i \in \{1, \dots, d\}$ be s.t. $\|v\|_\infty = |v_i| \neq 0$.

$$\Rightarrow \quad \|(I_d + M)v\|_\infty \geq |((I_d+M)v)_k| = \left| \sum_{j=1}^{d} (I_d+M)_{ij} v_j \right|$$

$$= \left| 1 + \sum_{j=1}^{d} M_{ij} \frac{v_j}{v_i} \right| \cdot |v_i| = \left| 1 + M_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^{d} \underbrace{M_{ij}}_{\leq 0} \underbrace{\frac{v_j}{v_i}}_{\in [-1,1]} \right| \cdot |v_i|$$

$$\geq \Big| 1 + \underbrace{M_{ii} + \sum_{j \neq i} M_{ij}}_{\geq 0} \Big| \cdot |v_i| \geq |v_i| = \|v\|_\infty .$$

ii) $$\|(I_d - M)v\|_\infty = \max_i \left| v_i - M_{ii} v_i - \sum_{j \neq i} M_{ij} v_j \right|$$

$$= \max_i \left| \underbrace{(1 - M_{ii})}_{\geq 0} v_i + \sum_{j \neq i} \underbrace{(-M_{ij})}_{\geq 0} v_j \right|$$

$$\leq \max_i \; (1 - M_{ii}) \|v\|_\infty + \sum_{j \neq i} (-M_{ij}) \|v\|_\infty = \max_i \|v\|_\infty \underbrace{- \sum_{i=1}^{d} M_{ij} \|v\|_\infty}_{\leq 0} \leq \|v\|_\infty$$

$\boxtimes$

Recall stability for $L_h u_h = f_h$, we called the scheme stable if $|||L_h^{-1}||| < c$ for some constant c which is independent of h

In other words: $\|u_h\|_h = \|L_h^{-1} f_h\|_h \leq c \|f_h\|_h$ so the discrete solution u_h should be bounded by a constant (indp. of h!) and the norm of the given data f_h

$\rightsquigarrow$ we will use this interpretation of stability and show that the discrete solution of the θ-scheme is bounded in terms of the given data, namely the initial condition u_0 and the right-hand side f

We can rewrite the θ-scheme (in vector form) as

$$u_h^{k+1} + \tau \theta L_h u_h^{k+1} = \tau f_h^k + (I - \tau(1-\theta) L_h) u_h^k \qquad (*)$$

We first apply estimate i) to the matrix $M = \tau \theta L_h$ (assumptions true!) and obtain

$$\|u_h^{k+1}\|_\infty \overset{i)}{\leq} \|(I + M) u_h^k\|_\infty \overset{(*)}{=} \|\tau f_h^k + (I - \tau(1-\theta) L_h) u_h^k\|_\infty$$

$$\leq \tau \|f_h^k\|_\infty + \|(\underbrace{I - \tau(1-\theta) L_h}_{= \tilde{M}}) u_h^k\|_\infty$$

IF $0 < \tilde{M}_{ii} \leq 1$, we can also apply estimate ii) of the theorem to obtain

$$\|u_h^{k+1}\|_\infty \leq \tau \|f_h^k\|_\infty + \|(I - \tilde{M}) u_h^k\|_\infty \overset{ii)}{\leq} \tau \|f_h^k\|_\infty + \|u_h^k\|_\infty$$

$$\underset{\text{as before}}{\overset{\text{again}}{\leq \dots \leq}} \underbrace{\|u_h^0\|_\infty}_{\text{initial data}} + \tau \sum_{j=1}^{k} \underbrace{\|f_h^k\|_\infty}_{\text{right hand side}} \Rightarrow \text{stability!}$$

When do we have $0 < \tilde{M}_{ii} \leq 1$? $0 < \tau (1-\theta) \frac{2}{h^2} \leq 1 \Rightarrow \underbrace{(1-\theta)\tau \leq \frac{h^2}{2}}_{\text{always true for } \theta = 1 \text{ !}}$

Altogether we now have shown (consistency + stability $\Rightarrow$ convergence) the following result:

<u>Theorem</u> Let $(1-\theta)\tau \leq \frac{h^2}{2}$ and $u \in C^{\overset{\text{time}}{2},\overset{\text{space}}{4}}$. Then

$$\max_k \|u_h^k - R_h u^k\|_\infty = \mathcal{O}(\tau + h^2).$$

If $\theta = \frac{1}{2}$ with $\tau \leq h^2$ and $u \in C^{3,4}$ then

$$\max_k \|u_h^k - R_h u^k\|_\infty = \mathcal{O}(\tau^2 + h^2).$$